

OLAP PRACTICAL 4

DATASET:

@relation sales

@attribute Region {North,South,East,West}

@attribute Product {Laptop,Mobile,Tablet}

@attribute Year numeric

@attribute Sales numeric

@data

North,Laptop,2022,50000

North,Mobile,2022,30000

South,Laptop,2022,45000

East,Tablet,2022,20000

West,Mobile,2022,25000

North,Laptop,2023,55000

South,Tablet,2023,22000

East,Mobile,2023,35000

West,Laptop,2023,48000

South,Mobile,2023,32000

Steps

1 Open WEKA

Launch WEKA software.

WEKA GUI Chooser window appears.

You will see options:

Explorer

Experimenter

Knowledge Flow

Simple CLI

Step 2: Open Explorer

Step 3: Load Dataset

Step 5: Perform Roll-Up Operation

What is Roll-Up?

Roll-Up summarizes detailed data into higher-level information.

Example:

Summarizing yearly sales region-wise.

Procedure for Roll-Up

Step 5.1:

Go to:

Preprocess Tab

Step 5.2:

Use filter:

Filter → Unsupervised → Attribute → Remove

step 5.3:

Remove Product attribute temporarily.

This groups data mainly by:

Region

Year

Step 5.4:

Observe summarized sales data.

This represents Roll-Up because detailed product-level information is reduced.

Step 6: Perform Drill-Down Operation

What is Drill-Down?

Drill-Down means moving from summarized data to detailed data.

Example:

- Viewing individual product sales inside each region.

Procedure for Drill-Down

Step 6.1:

Reload original dataset.

Click:

Open File

Load:

sales.arff

Step 6.2:

Now all attributes appear again:

- Region
- Product
- Year
- Sales

Step 6.3:

Select individual attributes one by one.

Observe:

- histograms,
- distributions,
- product-wise records.

This detailed analysis represents Drill-Down operation.

Step 7: Perform Slice Operation**What is Slice?**

Slice selects a single dimension value.

Example:

- Show only sales for Year = 2023.
-

Procedure for Slice**Step 7.1:**

Go to:

Filter

Step 7.2:

Choose:

Unsupervised → Instance → RemoveWithValues

Step 7.3:

Set options:

- Choose attribute:

Year

- Select:

2022

Remove 2022 records.

Step 7.4:

Click:

Apply

Result

Dataset now contains only:

Year = 2023

This is Slice operation.

Step 8: Perform Dice Operation

What is Dice?

Dice selects multiple dimensions together.

Example:

- Region = North
 - Product = Laptop
-

Procedure for Dice

Step 8.1:

Use:

RemoveWithValues Filter

multiple times.

Step 8.2:

Keep only:

- Region = North
- Product = Laptop

Remove all other combinations.

Step 8.3:

Click:

Apply

Result

Filtered multidimensional subset appears.

This is Dice operation.

Step 9: Perform Pivot Operation

What is Pivot?

Pivot rotates data view for better analysis.

Example:

- Compare Regions against Products.

Procedure for Pivot

WEKA does not directly support graphical pivot tables like spreadsheet software.

But pivot-like analysis can be done by:

- selecting different attributes,
- changing visualization axes,
- comparing histograms.

Step 9.1:

Go to:

Visualize Tab

Step 9.2:

Select:

- X-axis = Region
- Y-axis = Sales

Observe graph.

Step 9.3:

Now change:

- X-axis = Product
- Y-axis = Sales

This rotated analysis is similar to Pivot operation.

